Chakram Siri

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.00
Sri Chaitanya Junior Kalasala	Class XII (SSC)	2024	96.30%
Paramita Heritage School	Class X (CBSE)	2022	90.80%

PROJECTS

Book Recommendation System (GitHub | Live)

- Built a collaborative filtering-based book recommendation system using user-based and item-based similarity on 23,273 ratings from 2,212 users across 537 books.
- Implemented user-based, item-based, hybrid, and popularity-based recommendation models and compared their performance using Precision@5 and Recall@5.
- Achieved 0.0999 Recall@5 with User-Based CF, representing a 130.21% improvement over the popularity baseline.
- Developed and deployed a FastAPI web application supporting title-based book search, book cover images, and personalized recommendations.

Enhanced xv6 Operating System Kernel (GitHub)

- Extended the xv6 RISC-V kernel by implementing custom system calls, process accounting, and parent-child process management.
- Designed and implemented a System-Call-Aware Multi-Level Feedback Queue (MLFQ) scheduler with dynamic priority adjustment, starvation prevention, and per-process scheduling statistics.
- Developed a virtual memory subsystem featuring Clock page replacement, lazy allocation support, swapping, and per-process memory statistics for efficient memory management.
- Built a disk-backed swap subsystem with configurable FCFS/SSTF disk scheduling and simulated RAID 0/1/5 storage to improve swap management and fault tolerance.

PostgreSQL Query Optimization (GitHub)

- Analyzed PostgreSQL query execution plan transitions using the TPC-H benchmark under varying data selectivities.
- Automated execution plan analysis with Python (psycopg2), evaluating optimizer behavior by comparing forced and selected execution plans.
- Used pg_hint_plan to force alternative execution plans and evaluate PostgreSQL optimizer decisions using warm-cache execution times.
- Studied execution plan transitions involving Index Scan, Bitmap Scan, and Parallel Sequential Scan across multiple benchmark queries.

RISC-V FPU Register Optimization for Matrix Multiplication (GitHub)

- Implemented and optimized RISC-V assembly matrix multiplication by utilizing Floating Point Unit (FPU) registers for intermediate computations, minimizing memory accesses and improving register-level data reuse.
- Benchmarked the implementation using the gem5 architectural simulator, analyzing CPU cycles and instruction counts across 16×16, 32×32, 64×64, and 128×128 matrix sizes, achieving up to 1.63× speedup and approximately 6-7% fewer executed instructions.

SKILLS

Programming Languages: C, C++, Python, Java, SQL, Assembly (RISC-V)

Technical Skills: Linux, OOP, STL, PostgreSQL

Tools & Platforms: GitHub, VS Code, Google Colab, Autodesk Fusion

RELEVANT COURSES

Computer Science: Programming in C, Discrete Mathematics, Software Development Fundamentals, Data Structures, Algorithms, Theory of Computation, Operating Systems, Database Management Systems, Computer Architecture, Artificial Intelligence, Compilers-I

Mathematics: Calculus, Probability and Random Variables, Linear Algebra, Transform Techniques, Statistics

POSITIONS OF RESPONSIBILITY

Deputy Coordinator, Publicity -- Elan & nVision 2025

- Directly engaged students and club representatives across campus to drive event participation, tailoring outreach based on club interests and past engagement.
- Led and coordinated a team of volunteers, delegating outreach zones and aligning messaging to ensure consistent promotion of fest offerings.
- Acted as a point of contact between the publicity team and student clubs, balancing multiple stakeholder asks under fast deadlines.

EXTRACURRICULAR

NSS | IIT Hyderabad

- Participated in campus-level community drives including blood donation and cleanliness/awareness campaigns, engaging directly with both the student body and local community.
- Volunteered in tutoring/teaching sessions for underprivileged children, communicating with a non-technical, non-peer audience -- a different stakeholder skill than campus/technical work.